

Swin Transformer: Hierarchical Vision Transformer using Shifted Windows

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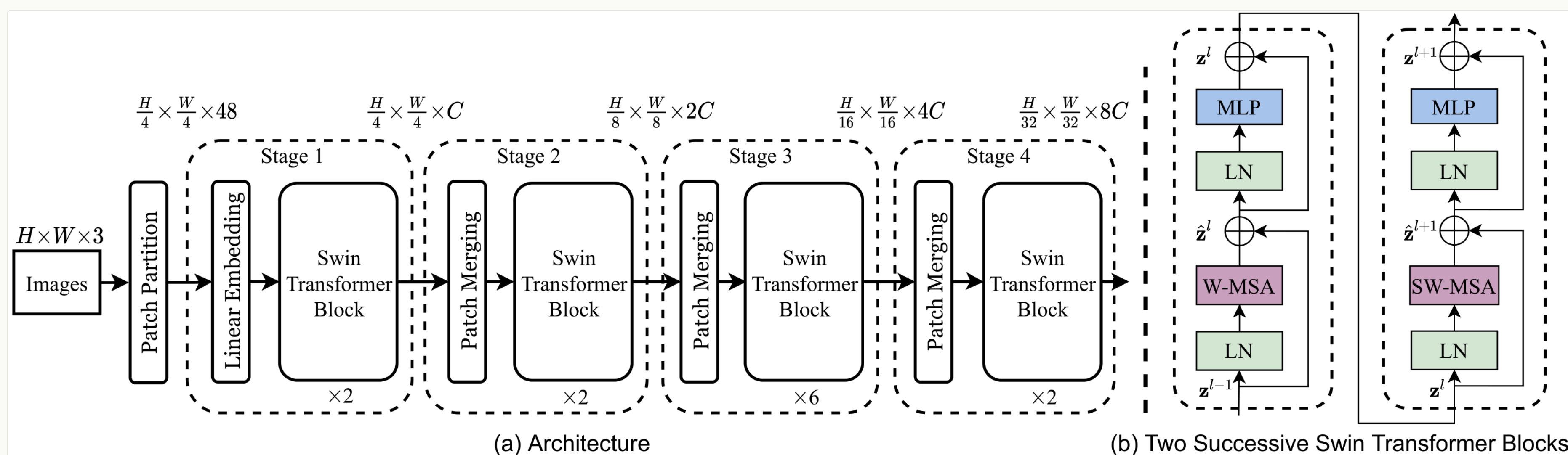
Paper



Code

Method

- Partition the image into **non-overlapping M×M windows** and compute self-attention only inside each window, giving **linear complexity** in image size instead of ViT's quadratic.
- Alternate consecutive blocks between a regular partition and one **shifted by $\lfloor M/2 \rfloor$** , so attention propagates across window boundaries every layer.
- Use **patch merging** between four stages to halve resolution and double channels, producing a **hierarchical pyramid** at H/4, H/8, H/16, H/32 — drop-in for ResNet backbones.



Swin-T architecture: 4 stages with patch merging between them; each block alternates W-MSA (window) and SW-MSA (shifted window) self-attention.

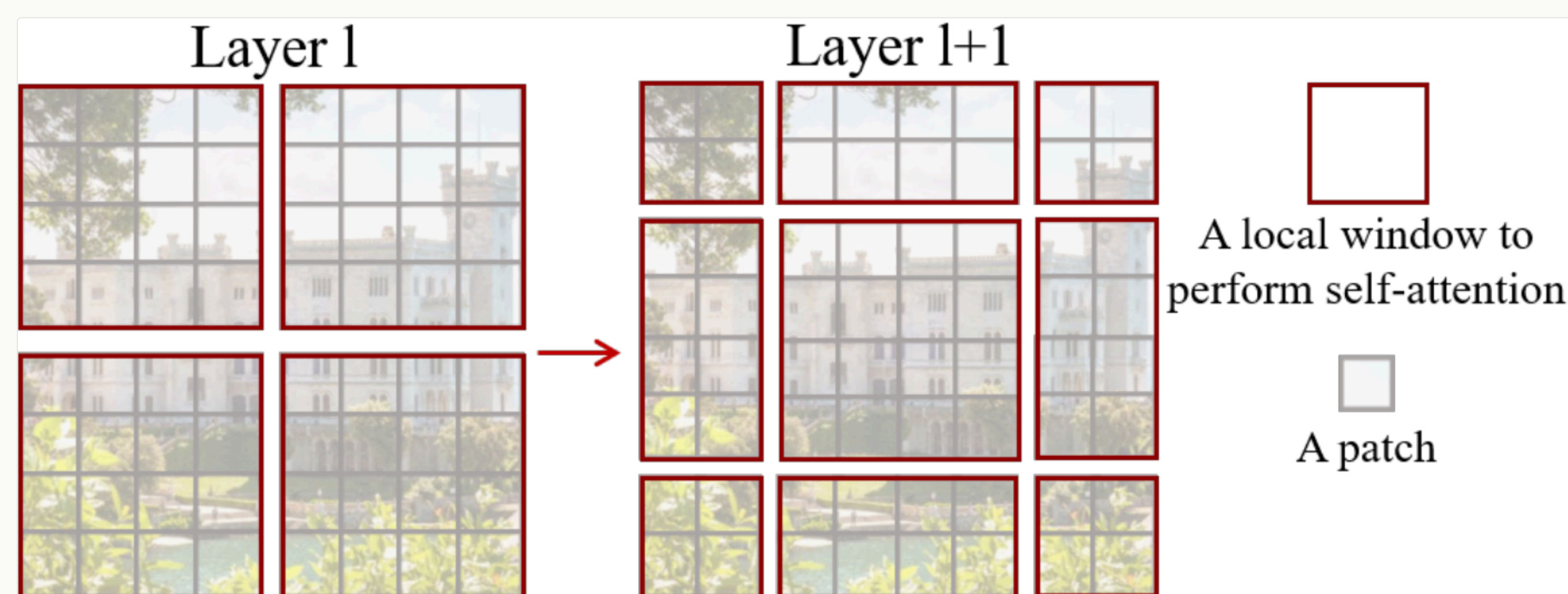
Problem

Standard vision Transformers like **ViT** compute **global self-attention** at a single low resolution, giving **quadratic cost** in image size and feature maps unsuitable as a general-purpose backbone for dense prediction tasks.

Visual elements vary enormously in scale and images carry **far higher token counts** than text, so per-pixel tasks like detection and segmentation are intractable for ViT-style global attention on high-resolution inputs.

Motivation

A general-purpose backbone needs **hierarchical multi-scale features** and **linear complexity** in image size. Plain non-overlapping windows give linear cost but **cut off information flow between windows**; alternating regular and shifted partitions fixes this without per-pixel sliding-window latency.



Takeaway

Shifting **non-overlapping local windows** between layers turns the Transformer into a **hierarchical, linear-complexity vision backbone** — SOTA across classification, detection, and segmentation, and a **drop-in alternative to ResNet** for dense prediction pipelines.

The same recipe also lifts all-MLP backbones, suggesting **hierarchy and local-window communication** are general design primitives for vision modeling beyond attention alone — Swin is now the default Transformer backbone for high-resolution dense prediction.

Dataset / Benchmark

- ImageNet-1K** (1.28M train / 50K val / 1000 classes) and **ImageNet-22K** (14.2M images / 22K classes) for classification.
- COCO 2017** (118K train / 5K val / 20K test-dev) for detection & instance segmentation; **ADE20K** (150 classes) for semantic segmentation.

Key Results

Task / Metric	Prior SOTA	Swin-L
ImageNet-1K top-1 (IN-22K)	86.4	87.3
COCO box AP (test-dev)	56.0	58.7
COCO mask AP (test-dev)	48.5	51.1
ADE20K mIoU (val)	50.3	53.5

+2.7 box AP · +2.6 mask AP · +3.2 mIoU vs. previous SOTA

SO WHAT → A single backbone family sets new SOTA across all three tasks at unusually large margins; with ImageNet-1K only training, **Swin-T (81.3%)** still matches or beats DeiT, RegNet and EfficientNet.

Ablation Study

- Removing the **shifted-window scheme** drops **1.1%** ImageNet, **2.8** box AP / **2.2** mask AP COCO, **2.8** mIoU ADE20K.
- Learned **relative position bias** beats no positional encoding and absolute position embeddings across all three tasks.
- Cyclic-shift keeps shifted attention as fast as plain windowed (3.0 vs 2.8 ms), **~40× faster** than naive sliding-window.

Headline Numbers

87.3%

58.7

COCO box AP

51.1

COCO mask AP

53.5

ADE20K mIoU